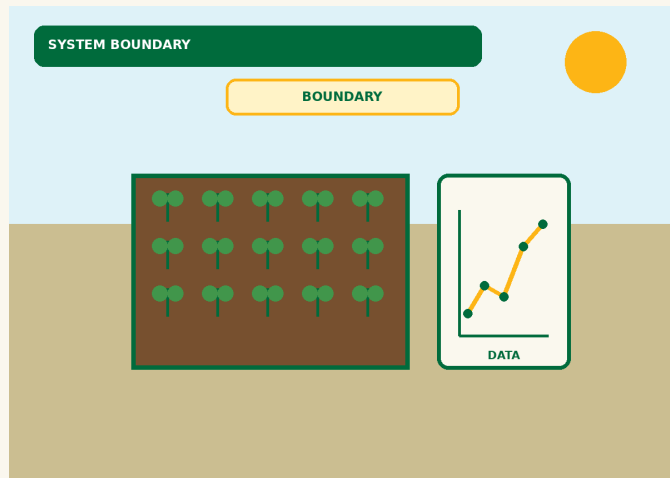




BOUNDARY

SYSTEM BOUNDARY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 781 • SYSTEM BOUNDARY

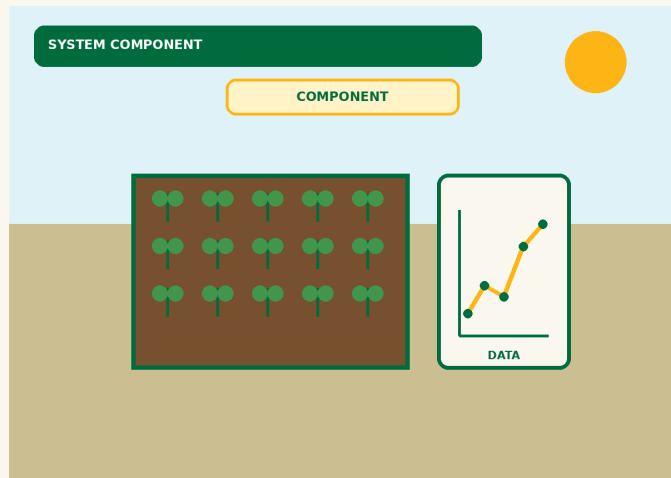
CONCEPT

A system boundary defines what components and processes are included in an analysis and what lies outside it.

INVESTIGATION TASK

Draw a boundary around the school garden system.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



COMPONENT

SYSTEM COMPONENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 782 • SYSTEM COMPONENT

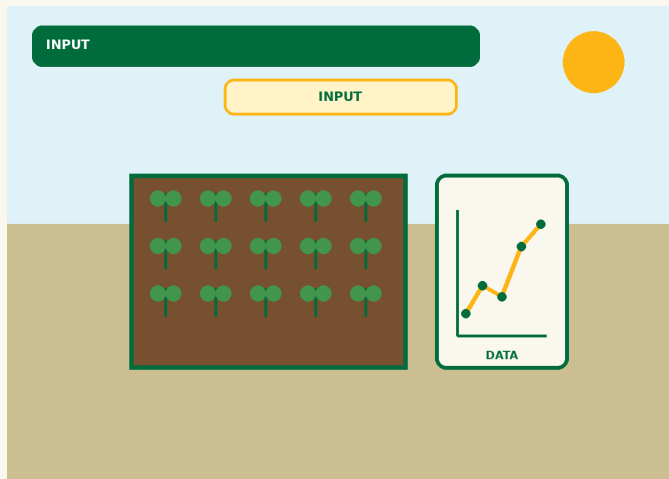
CONCEPT

Components are interacting physical, biological, informational, and human elements within the system.

INVESTIGATION TASK

Classify the main garden components.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INPUT

INPUT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

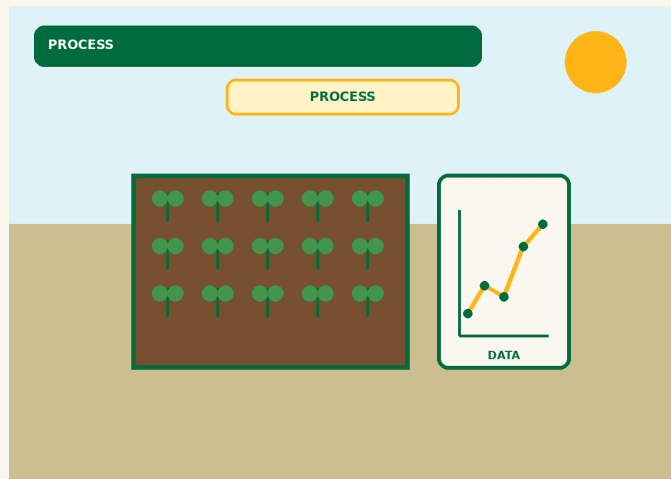
CONCEPT

System inputs are resources or information entering the defined boundary.

INVESTIGATION TASK

Quantify one input during a production activity.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PROCESS

PROCESS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 784 • PROCESS

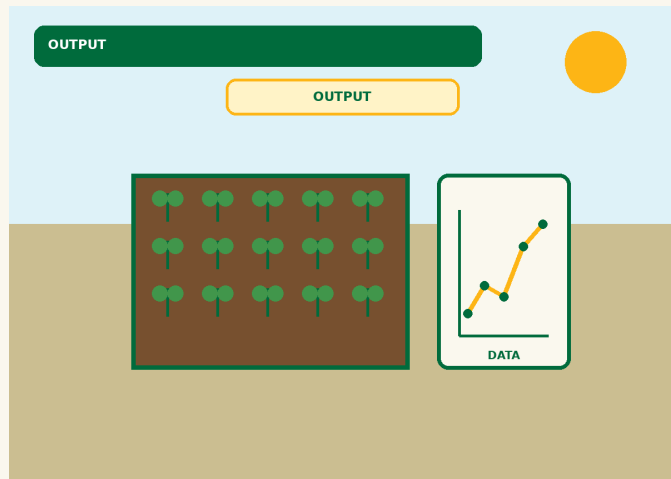
CONCEPT

Processes transform, move, store, or regulate materials, energy, information, and biological resources.

INVESTIGATION TASK

Identify three processes in the garden.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



OUTPUT

OUTPUT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 785 • OUTPUT

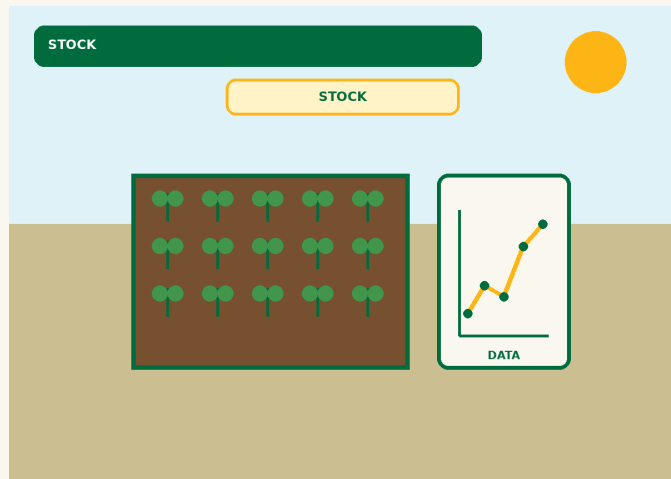
CONCEPT

Outputs are products, wastes, measurements, or effects leaving the system boundary.

INVESTIGATION TASK

Classify useful and unwanted outputs.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



STOCK

STOCK

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

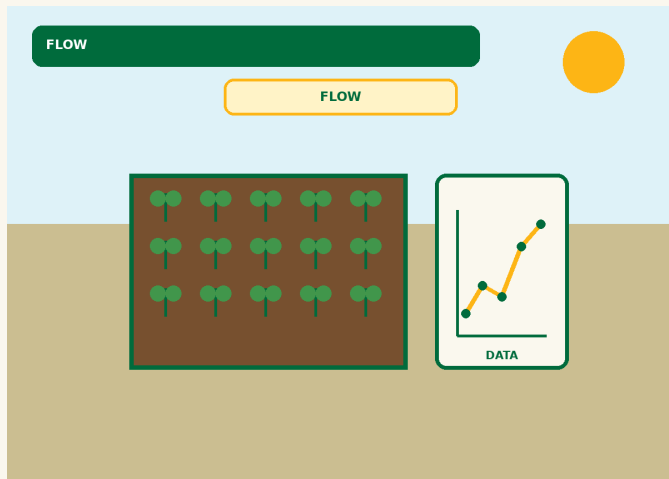
CONCEPT

A stock is a quantity stored within the system, such as water in a tank, nutrients in soil, or biomass in crops.

INVESTIGATION TASK

Measure or estimate one stock.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FLOW

FLOW

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 787 • FLOW

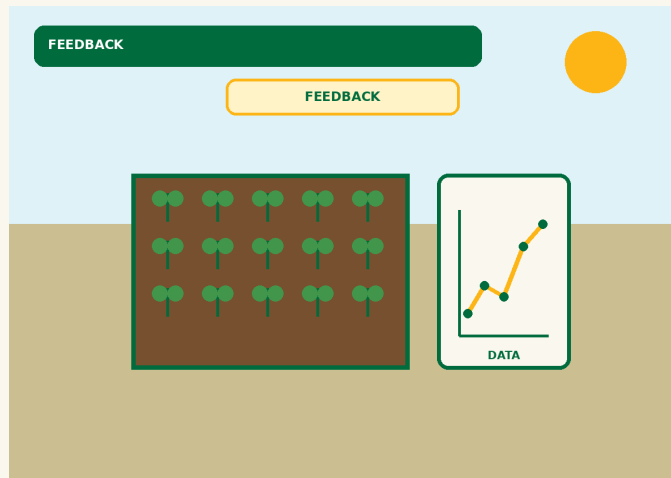
CONCEPT

A flow is the rate at which a material or resource enters, leaves, or moves between stocks.

INVESTIGATION TASK

Measure or calculate one flow.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FEEDBACK

FEEDBACK

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

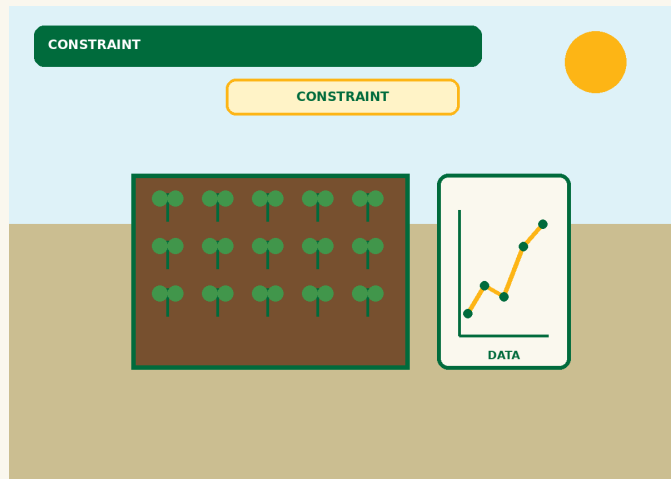
CONCEPT

Feedback occurs when information about system condition influences a later action or process.

INVESTIGATION TASK

Identify a monitoring-to-management feedback loop.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CONSTRAINT

CONSTRAINT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 789 • CONSTRAINT

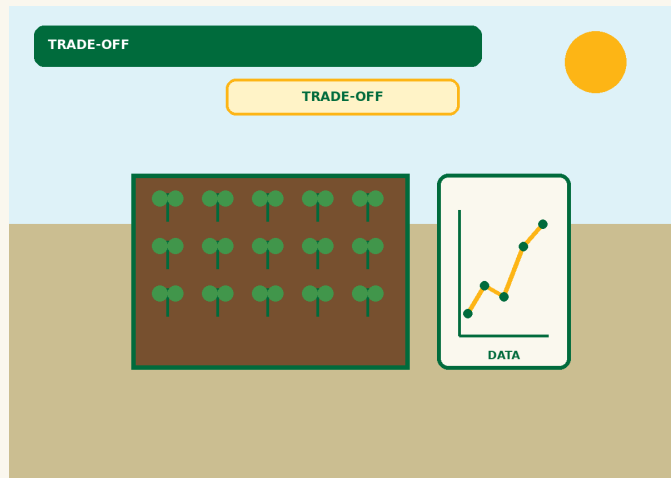
CONCEPT

A constraint limits system performance, choices, or resource availability.

INVESTIGATION TASK

Identify the current limiting constraint.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TRADE-OFF

TRADE-OFF

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 790 • TRADE-OFF

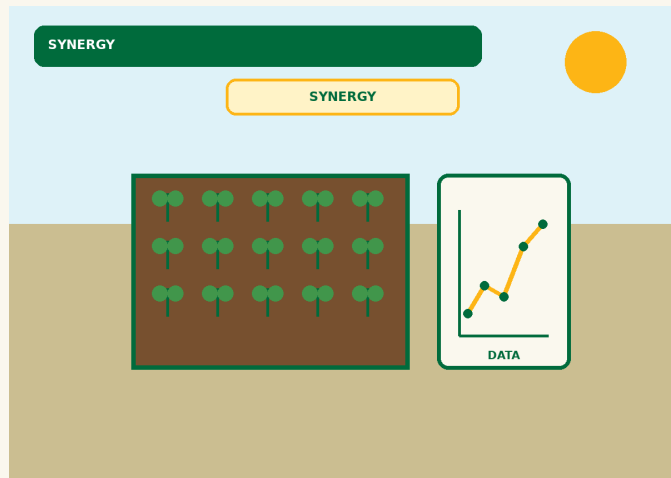
CONCEPT

A trade-off occurs when improving one objective can reduce performance in another.

INVESTIGATION TASK

Describe one garden-management trade-off.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SYNERGY

SYNERGY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

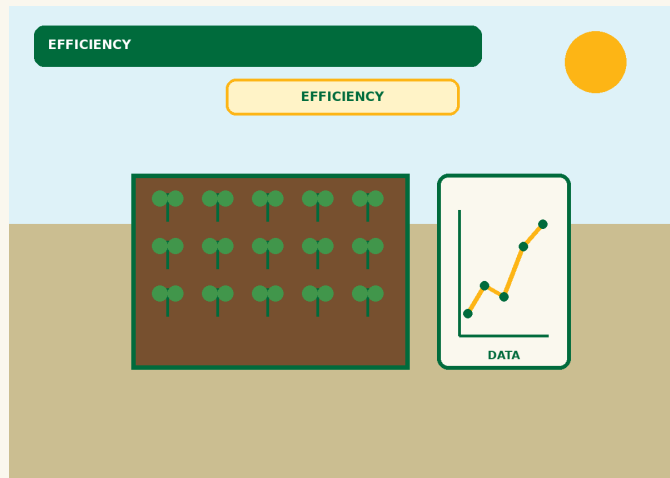
CONCEPT

A synergy occurs when one action produces benefits for multiple system objectives.

INVESTIGATION TASK

Identify one action with more than one benefit.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EFFICIENCY

EFFICIENCY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 792 • EFFICIENCY

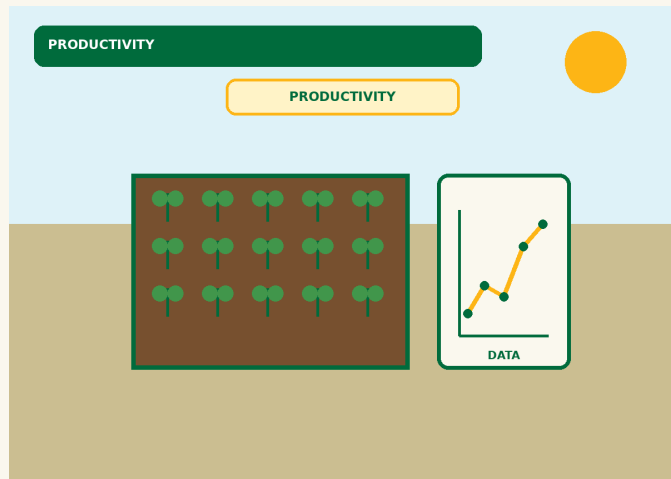
CONCEPT

Efficiency compares useful output with the resources supplied or losses avoided under a defined metric.

INVESTIGATION TASK

Calculate a simple system efficiency.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PRODUCTIVITY

PRODUCTIVITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 793 • PRODUCTIVITY

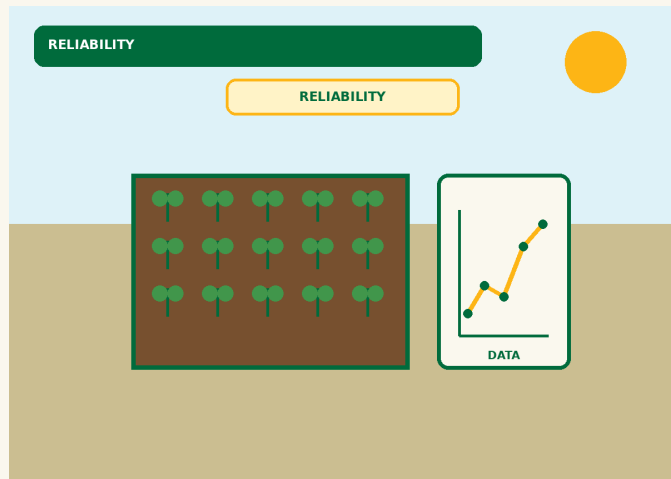
CONCEPT

Productivity relates useful production to land, water, labor, time, or another resource basis.

INVESTIGATION TASK

Calculate one productivity indicator.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RELIABILITY

RELIABILITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 794 • RELIABILITY

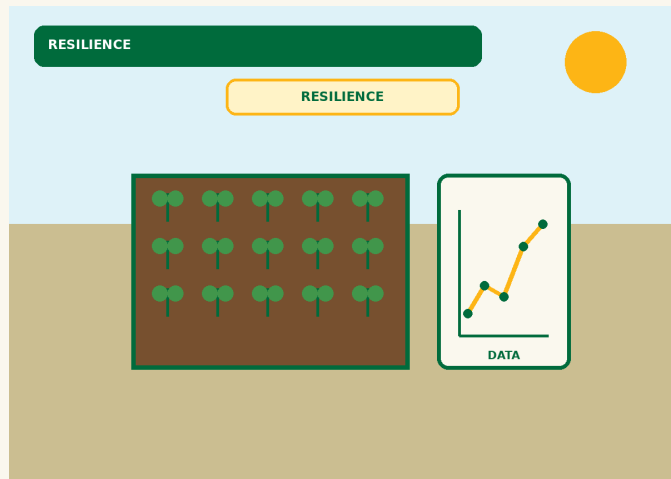
CONCEPT

Reliability describes the ability of a system to perform its required function consistently over time.

INVESTIGATION TASK

Use records to identify a reliability problem.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RESILIENCE

RESILIENCE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 795 • RESILIENCE

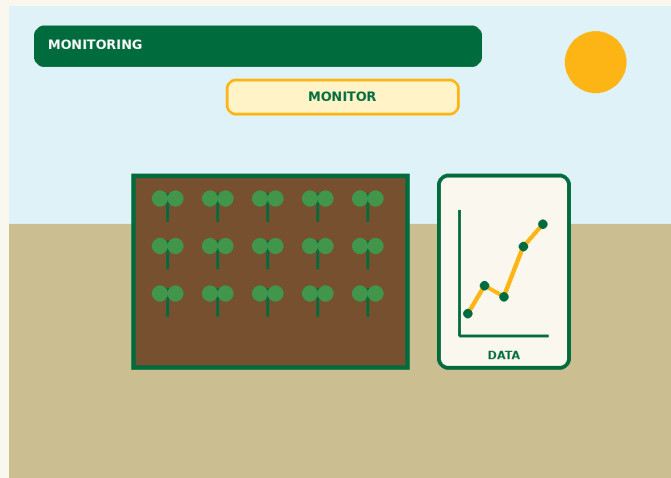
CONCEPT

Resilience is the capacity of a system to withstand, adapt to, and recover from disturbances while maintaining important functions.

INVESTIGATION TASK

Evaluate a response to a water-supply interruption.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MONITOR

MONITORING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

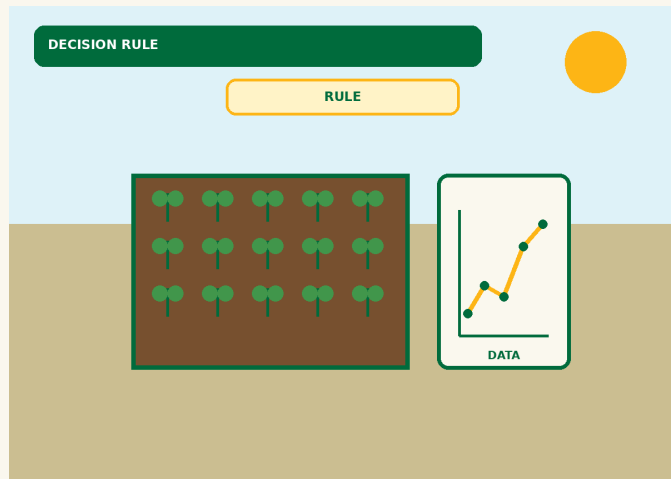
CONCEPT

Monitoring provides observations and measurements needed to understand current system condition.

INVESTIGATION TASK

Select indicators for a management dashboard.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RULE

DECISION RULE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 797 • DECISION RULE

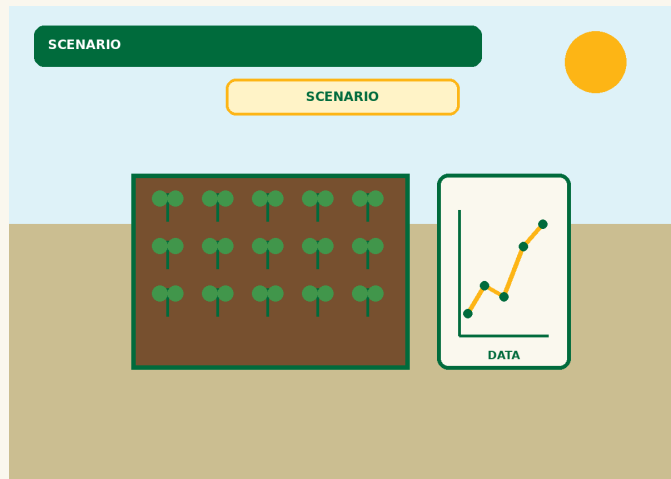
CONCEPT

A decision rule converts measurements or observations into a defined management action.

INVESTIGATION TASK

Write a threshold-based irrigation rule.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SCENARIO

SCENARIO

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 798 • SCENARIO

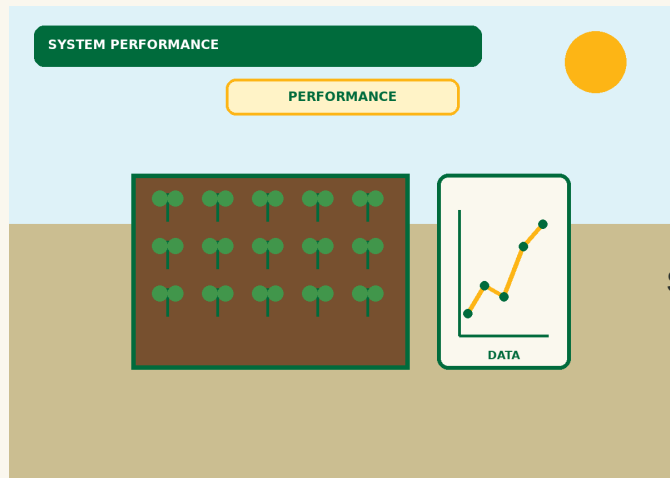
CONCEPT

A scenario is a structured description of possible conditions used to test how a system may perform.

INVESTIGATION TASK

Compare normal and resource-constrained scenarios.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PERFORMANCE

SYSTEM PERFORMANCE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 799 • SYSTEM PERFORMANCE

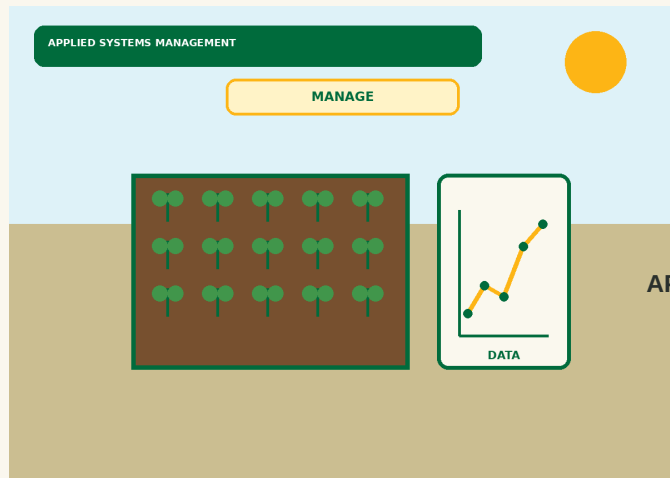
CONCEPT

System performance is evaluated with multiple indicators because no single metric captures all objectives.

INVESTIGATION TASK

Select indicators for production, water, and reliability.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MANAGE

APPLIED SYSTEMS MANAGEMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CONCEPT

Applied systems management uses boundaries, measurements, feedback, trade-offs, scenarios, and performance indicators to guide decisions.

INVESTIGATION TASK

Propose and justify one integrated management improvement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE